Two-Block Circuit Competition with ReLU(tanh) Neurons

Summary. We repeat the fraction-aware two-block derivation for $\phi(z) = \mathrm{ReLU}(\tanh z) = [\tanh z]_+$. The steps and notation match the pure-ReLU theory, but saturation removes positive homogeneity. Consequently, neuron counts no longer cancel from the weight-decay cost, the lower-mode leakage ratios depend on the neuron amplitude, and direct and compositional block-2 circuits can coexist when their fractions are constrained. The familiar accessibility boundary $P/S = 1/\sqrt{8}$ survives only as the weak-activation onset criterion. We show both the phase diagram conditional on fixed fractions and the result of minimizing the same free energy over the fractions themselves.

1. Model, circuit populations, and assumptions

We keep exactly the objective and residual architecture of the ReLU report:

$$\begin{aligned} &h_1 = h_0 + \phi(V_1 h_0), \qquad h_2 = h_1 + \phi(V_2 h_1), \\ &\mathcal{J} = \frac{\gamma_2}{2} \sum_{a\in\{12,34\}} (1-\widehat{A}_a)^2 + \frac{\gamma_4}{2}(1-\widehat{A}_4)^2 + \frac{1}{2}\big(\lambda_E\|E\|^2 + \lambda_1\|V_1\|^2 + \lambda_2\|V_2\|^2 \\ &\qquad\qquad +\lambda_p\|W_{d2}\|^2 + \lambda_4\|w_4\|^2\big)\,, \end{aligned} \tag{1}$$

where $\widehat{A}_a = \mathbb{E}[f_a\chi_a]$. Since $\chi_a^2 = 1$, the aligned pure-mode loss is $(1-\widehat{A}_a)^2$ for either target sign. The available populations are

$$\begin{aligned} &m_{12} = N_1\rho^{(1)}_{2\times1}, \qquad m_{14} = N_1\rho^{(1)}_{4\times1}, \\ &m_{22} = N_2\rho^{(2)}_{2\times1}, \qquad m_D = N_2\rho^{(2)}_{4\times1}, \qquad m_C = N_2\rho^{(2)}_{2\times2}, \end{aligned} \tag{2}$$

with the unlisted fraction idle or untracked. The 2×1 populations are split equally between the two canonical parent pairs.

For a balanced neuron with p binary parent modes $y_i \in \{\pm1\}$, write $z = \xi\sum_{i=1}^p y_i$. Every Walsh coefficient is exactly

$$c_A(\xi) = 2^{-p} \sum_{y\in\{\pm1\}^p} \left[\tanh\left(\xi\sum_i y_i\right)\right]_+ \prod_{i\in A} y_i. \tag{3}$$

For a two-input gate, the constant, two parents, and product all have magnitude

$$a(\xi) = \frac{\tanh(2\xi)}{4}. \tag{4}$$

For a four-input gate, put $t_j = \tanh(j\xi)$. The coefficients at degrees $0,1,2,3,4$ are, respectively,

$$b_0 = \frac{t_4+4t_2}{16}, \quad b_1 = \frac{t_4+2t_2}{16}, \quad b_2 = \frac{t_4}{16}, \quad b_3 = \frac{t_4-2t_2}{16}, \quad -b_4 = \frac{t_4-4t_2}{16}, \tag{5}$$

where $b_4 = (4t_2 - t_4)/16 > 0$ denotes the target magnitude. Equations (4)–(5) were checked by direct enumeration of the 2^p inputs. As $\xi\to0$, $(a, b_1, b_2, b_4) \sim (\xi/2, \xi/2, \xi/4, \xi/4)$, reproducing the ReLU ratios. As $\xi\to\infty$, they approach $(1/4, 3/16, 1/16, 3/16)$: the ratios are no longer fixed.

We retain the ReLU report's orthogonal-mode, orthogonal-population, sign-balance, aligned-readout, sufficient-width, and global-minimum assumptions. The additional closure is (A8): neurons in one listed population are exchangeable and have a common ξ. This is exact for a constrained homogeneous population and gives the simplest finite-fraction theory. If amplitudes within a four-input population are unrestricted, their distribution is an additional order parameter; the final paragraph states the exact replacement.

2. What changes in the fraction step

Let every parent coefficient vector have norm R. A balanced p-input neuron with preactivation amplitude ξ has incoming norm $\sqrt{p}\,\xi/R$. Thus m identical neurons producing coefficient magnitude $c(\xi)$ obey

$$Q^2 = mc(\xi)^2, \qquad \frac{\lambda}{2}\sum_{j=1}^m \|v_j\|^2 = \frac{\lambda mp}{2R^2}\xi^2. \tag{6}$$

For a two-input population, $a^{-1}(u) = \frac{1}{2}\operatorname{arctanh}(4u)$, so eliminating ξ gives

$$\boxed{\mathcal{C}_{2,m}(Q;R) = \frac{\lambda m}{4R^2}\operatorname{arctanh}^2\!\left(\frac{4Q}{\sqrt{m}}\right)}, \qquad Q < \frac{\sqrt{m}}{4}. \tag{7}$$

For a four-input target population,

$$\boxed{\mathcal{C}_{4,m}(Q;R) = \frac{2\lambda m}{R^2}\left[b_4^{-1}\!\left(\frac{Q}{\sqrt{m}}\right)\right]^2}, \qquad Q < \frac{3\sqrt{m}}{16}, \tag{8}$$

where the monotone inverse is evaluated numerically. Unlike the quadratic ReLU costs, Eqs. (7)–(8) depend on $m = N\rho$ and diverge at the saturation capacity. Expanding arctanh and b_4^{-1} at small $Q/\sqrt{m}$ recovers the fraction-independent ReLU costs. This identifies the physical distinction: changing a positive fraction does nothing in a homogeneous activation, but changes both marginal cost and capacity once the activation saturates.

The ReLU winner-take-all target-strength result also changes. With a fixed total block-2 target norm, putting everything into the cheaper linear route was optimal. Here each route's marginal cost rises toward its capacity, so the initially cheaper route can be joined by the other route. The relevant observable remains the target-strength share

$$\alpha_C \equiv \frac{Q_C^2}{Q_D^2 + Q_C^2}, \tag{9}$$

but α_C can now lie strictly between zero and one at $T = 0$.

3. Coupled mode geometry and reduced free energy

Following the ReLU derivation, retain s for the residual input-singleton norm. Replace its four learned strengths by five population amplitudes $\xi_{12}, \xi_{14}, \xi_{22}, \xi_D, \xi_C$. The corresponding target strengths are

$$\begin{aligned} q &= \sqrt{m_{12}/2}\, a(\xi_{12}), & r &= \sqrt{m_{14}}\, b_4(\xi_{14}), & p &= \sqrt{m_{22}/2}\, a(\xi_{22}), \\ Q_D &= \sqrt{m_D}\, b_4(\xi_D), & Q_C &= \sqrt{m_C}\, a(\xi_C). \end{aligned} \tag{10}$$

The new feature is amplitude-dependent leakage. Define

$$L_1 = \sqrt{m_{14}}\, b_1(\xi_{14}), \qquad L_2 = \sqrt{m_{14}}\, b_2(\xi_{14}), \qquad L_D = \sqrt{m_D}\, b_2(\xi_D). \tag{11}$$

The input singleton norm S, canonical-parent norm P, final parent norm Z, and final target norm H are then

$$\boxed{S^2 = s^2 + q^2 + L_1^2, \quad P^2 = q^2 + L_2^2, \quad Z^2 = P^2 + p^2 + L_D^2 + Q_C^2, \quad H^2 = r^2 + Q_D^2 + Q_C^2.} \tag{12}$$

This is the direct analogue of $(S^2, P^2) = (s^2 + q^2 + 4r^2, q^2 + r^2)$, now with the correct nonlinear ratios. Four cross-pair $d2$ modes and the $d3$ modes from the block-1 four-input population are also present, with norms L_2 and $\sqrt{m_{14}}|b_3(\xi_{14})|$; under the orthogonal-readout closure they enter neither supervised readout and have no downstream block.

Minimizing the two readouts at fixed hidden representation is unchanged:

$$B^* = \frac{\gamma_2 Z^2}{\gamma_2 Z^2 + \lambda_p}, \qquad A^* = \frac{\gamma_4 H^2}{\gamma_4 H^2 + \lambda_4}. \tag{13}$$

Substitution into Eq. (1) gives the single joint free energy

$$\begin{aligned} F = \min_{s,\,\xi_{12},\xi_{14},\xi_{22},\xi_D,\xi_C \geq 0} \Bigg\{ & 2\lambda_E s^2 + \frac{\lambda_1}{s^2}\left(m_{12}\xi_{12}^2 + 2m_{14}\xi_{14}^2\right) \\ & + \lambda_2\left(\frac{m_{22}\xi_{22}^2 + 2m_D\xi_D^2}{S^2} + \frac{m_C\xi_C^2}{P^2}\right) + \frac{\gamma_2\lambda_p}{\gamma_2 Z^2 + \lambda_p} + \frac{\gamma_4\lambda_4}{2(\gamma_4 H^2 + \lambda_4)}\Bigg\}. \end{aligned} \tag{14}$$

All quantities in Eq. (14) are defined by Eqs. (4), (5), and (10)–(12). This is one simultaneous minimization; its written order is only algebraic convenience. It includes both block-2 target routes at once, so it does not presume the answer to the circuit competition.

As a check, use the small-ξ limits in Eq. (14). The first-block costs become $8\lambda_1 q^2/s^2 + 32\lambda_1 r^2/s^2$, the block-2 parent cost becomes $8\lambda_2 p^2/S^2$, and the two target costs become

$$32\lambda_2\frac{Q_D^2}{S^2} + 4\lambda_2\frac{Q_C^2}{P^2} = \frac{1}{2}\left(\kappa_D Q_D^2 + \kappa_C Q_C^2\right), \qquad \kappa_D = \frac{64\lambda_2}{S^2}, \quad \kappa_C = \frac{8\lambda_2}{P^2}. \tag{15}$$

Thus the derivation reduces term-by-term to the previous ReLU theory.

4. Phase diagram and control implications

At infinitesimal target amplitude, composition first becomes cheaper when

$$\boxed{\kappa_C < \kappa_D \quad \Longleftrightarrow \quad P/S > 1/\sqrt{8}.} \tag{16}$$

At finite amplitude Eq. (16) is not the phase boundary: $b_1 : b_2 : b_4$ changes with ξ, the two route costs have different curvature and capacities, and their parent byproducts change the rest of the objective. The phase is instead read from the minimizer of Eq. (14): C-dominant if $\alpha_C > 0.9$, D-dominant if $\alpha_C < 0.1$, and mixed otherwise, with separate block-1-only, parent-only, and suppressed regions.

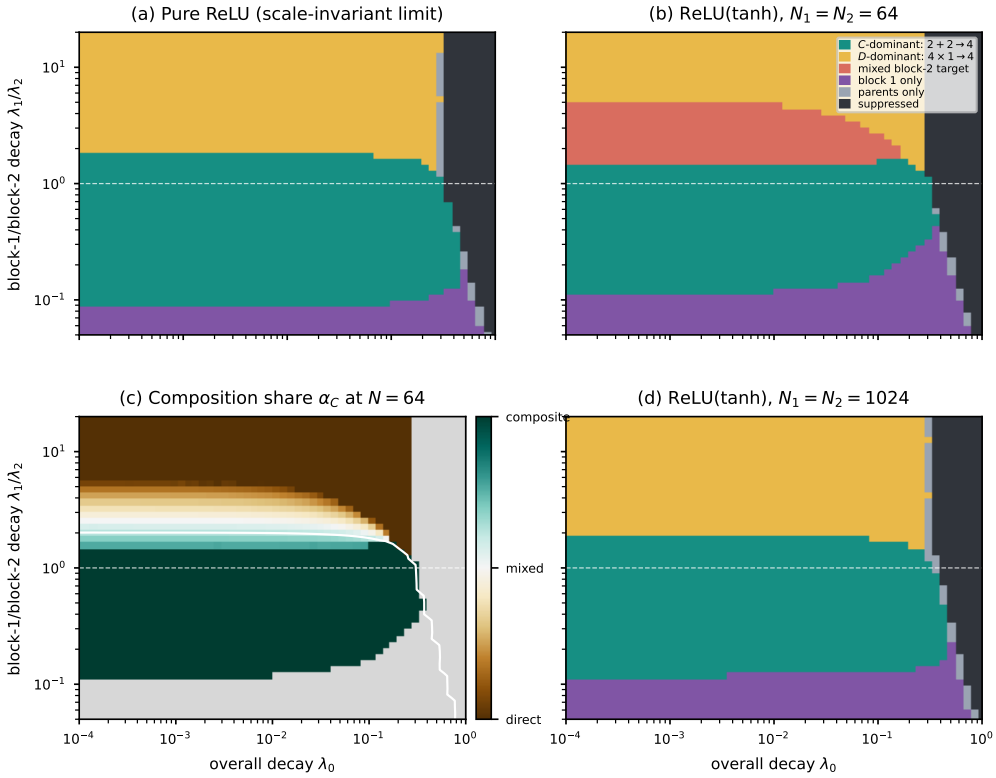


Figure 1: **Fixed-fraction homogeneous-population phase diagrams.** $\gamma_2 = \gamma_4 = 2$ and $\lambda_E = \lambda_p = \lambda_4 = \lambda_2 = \lambda_0$; the vertical coordinate is λ_1/λ_2. Panels (b)–(d) use $(\rho^{(1)}_{2\times1}, \rho^{(1)}_{4\times1}) = (1/2, 3/8)$ and $(\rho^{(2)}_{2\times1}, \rho^{(2)}_{4\times1}, \rho^{(2)}_{2\times2}) = (1/16, 1/8, 1/4)$. (a) Pure-ReLU reference. (b) At finite width, saturation produces a genuine mixed strip and moves the direct/compositional crossover. (c) Composition share; gray means no active block-2 target. The white line is $P/S = 1/\sqrt{8}$ and is only the weak-signal diagnostic. (d) Multiplying every population size by 16 moves each neuron into its linear regime, shrinks the mixed strip, and approaches panel (a). Each panel is a multistart numerical minimization of the displayed reduced free energy.

Minimizing over the fractions. Within A8, the fractions are ordinary additional order parameters. Define the two allocation simplices

$$\Delta_1 = \{(\rho_{12}, \rho_{14}) \colon \rho_{12}, \rho_{14} \geq 0,\ \rho_{12} + \rho_{14} \leq 1\}, \qquad \Delta_2 = \{(\rho_{22}, \rho_D, \rho_C) \colon \rho_c \geq 0,\ \rho_{22} + \rho_D + \rho_C \leq 1\}. \tag{17}$$

The unused mass is idle. The fully optimized homogeneous theory is simply

$$\boxed{F_{\text{opt}} = \min_{\rho^{(1)} \in \Delta_1,\, \rho^{(2)} \in \Delta_2} F\big(\rho^{(1)}, \rho^{(2)}\big),} \tag{18}$$

where F is the simultaneous amplitude minimization in Eq. (14) and $m_{\ell c} = N_\ell \rho^{(\ell)}_c$. Thus the plotted calculation has only eleven scalar order parameters: s, five amplitudes, and five independent fractions. We solve them jointly, not by first optimizing F at arbitrarily fixed features. A fraction assigned to a population with $\xi_c = 0$ is physically unidentifiable; in Fig. 2 such mass is absorbed into the idle category. For numerical display, a population is called active only if its propagated modes contribute at least 0.5% of the relevant retained mode norm.

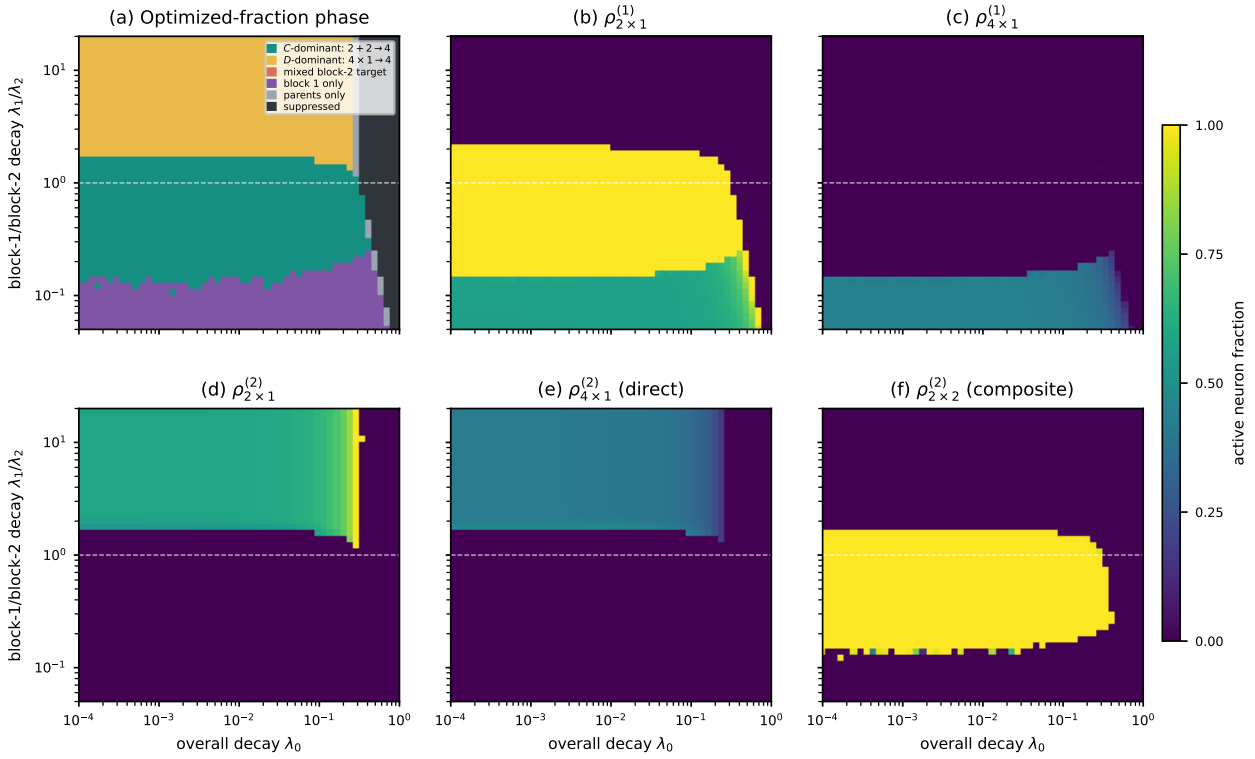


Figure 2: **Phase and active fractions after minimizing Eq.** (18)**.** $N_1 = N_2 = 64$ and all loss weights and decay conventions match Fig. 1. (a) The optimized phase. (b)–(c) Active first-block fractions. (d)–(f) Active second-block fractions. Unshown mass is idle. In this slice the optimizer finds no stable mixed block-2 target phase: allowing the allocation to move relieves the fixed-capacity saturation that made the mixed strip in Fig. 1(b). The diagrams are multistart numerical global-search approximations to the displayed eleven-dimensional free energy.

Away from phase boundaries the allocation has a simple interpretation. In the compositional phase the numerical median is $(\rho^{(1)}_{2\times1}, \rho^{(1)}_{4\times1}) \simeq (1, 0)$ and $(\rho^{(2)}_{2\times1}, \rho^{(2)}_{4\times1}, \rho^{(2)}_{2\times2}) \simeq (0, 0, 1)$. In the high-$\lambda_1/\lambda_2$ direct phase block 1 is inactive and block 2 splits as $(0.59, 0.41, 0)$ between parent and direct-target neurons. In the block-1-only phase the first block splits as $(0.565, 0.435)$ and block 2 is idle. These are energy-minimizing active fractions within A8, rather than fitted inputs or predictions of training dynamics.

How decay controls the circuit. Increasing λ_1/λ_2 suppresses the block-1 parent representation and favors the direct 4×1 route; making block 1 sufficiently cheap instead lets it solve $d4$ itself. Between these regimes, preserving P/S favors $2 + 2 \to 4$. Unlike pure ReLU, the result also depends on the available counts: increasing m_C raises the compositional capacity and lowers its finite-amplitude marginal cost, while increasing m_D does the same for the direct route. Therefore no fraction-independent finite-amplitude boundary exists when fractions are externally constrained. If neurons can freely change type, Eq. (18) predicts the allocation and, in the displayed slice, removes the mixed strip. In the large-width or weak-activation limit, Eq. (16) and the sharp pure-ReLU diagram are recovered.

Finally, A8 is a closure rather than a consequence of the activation. Without it, replace each common amplitude by a probability measure $\nu_c(d\xi)$ and replace, for example, $m_c b_k(\xi_c)^2$ by $m_c \int b_k(\xi)^2 \nu_c(d\xi)$ and $m_c \xi_c^2$ by $m_c \int \xi^2 \nu_c(d\xi)$. Minimizing Eq. (14) over these measures is the exact zero-temperature population theory. Equal sharing is optimal for the two-input cost in Eq. (7); a four-input population can instead split into inactive and finite-amplitude subpopulations because $[b_4^{-1}(\sqrt{x})]^2$ is not globally convex. This refinement can change the numerical fractions and phase boundaries. The robust conclusions are that saturation makes active fractions physical, constrained capacities can permit mixed block-2 solutions, and the ReLU boundary is restored as $N\rho \to \infty$ at fixed aggregate mode norm.